

Name : Dheeraj Vanmore
Roll no. : 24102B2004
Subject : R Programming Assignment 1

Github link : [DheerajVanmore/R-Assignments](https://github.com/DheerajVanmore/R-Assignments)

Task 1 : Import and Inspect the Dataset

```
===== TASK 1: Import and Inspect =====

--- First six records ---
  No year month day hour PM2.5 PM10 SO2 NO2 CO O3 TEMP PRES DEWP RAIN wd
1  1 2013     3   1   0     4     4   4   7 300 77 -0.7 1023.0 -18.8  0 NNW
2  2 2013     3   1   1     8     8   4   7 300 77 -1.1 1023.2 -18.2  0  N
3  3 2013     3   1   2     7     7   5  10 300 73 -1.1 1023.5 -18.2  0 NNW
4  4 2013     3   1   3     6     6  11  11 300 72 -1.4 1024.5 -19.4  0  NW
5  5 2013     3   1   4     3     3  12  12 300 72 -2.0 1025.2 -19.5  0  N
6  6 2013     3   1   5     5     5  18  18 400 66 -2.2 1025.6 -19.6  0  N
WSPM      station
1  4.4 Aotizhongxin
2  4.7 Aotizhongxin
3  5.6 Aotizhongxin
4  3.1 Aotizhongxin
5  2.0 Aotizhongxin
6  3.7 Aotizhongxin

--- Structure of dataset ---
'data.frame':   35064 obs. of  18 variables:
 $ No      : int  1 2 3 4 5 6 7 8 9 10 ...
 $ year    : int  2013 2013 2013 2013 2013 2013 2013 2013 2013 2013 ...
 $ month   : int  3 3 3 3 3 3 3 3 3 3 ...
 $ day     : int  1 1 1 1 1 1 1 1 1 1 ...
 $ hour    : int  0 1 2 3 4 5 6 7 8 9 ...
 $ PM2.5   : num  4 8 7 6 3 5 3 3 3 3 ...
 $ PM10    : num  4 8 7 6 3 5 3 6 6 8 ...
 $ SO2     : num  4 4 5 11 12 18 18 19 16 12 ...
 $ NO2     : num  7 7 10 11 12 18 32 41 43 28 ...
 $ CO      : int  300 300 300 300 300 400 500 500 500 400 ...
 $ O3      : num  77 77 73 72 72 66 50 43 45 59 ...
 $ TEMP    : num  -0.7 -1.1 -1.1 -1.4 -2 -2.2 -2.6 -1.6 0.1 1.2 ...
 $ PRES    : num  1023 1023 1024 1024 1025 ...
 $ DEWP    : num  -18.8 -18.2 -18.2 -19.4 -19.5 -19.6 -19.1 -19.1 -19.2 -19.3 ...
 $ RAIN    : num  0 0 0 0 0 0 0 0 0 0 ...
 $ wd      : chr  "NNW" "N" "NNW" "NW" ...
 $ WSPM    : num  4.4 4.7 5.6 3.1 2 3.7 2.5 3.8 4.1 2.6 ...
 $ station : chr  "Aotizhongxin" "Aotizhongxin" "Aotizhongxin" "Aotizhongxin" ...

--- Dimensions ---
Rows: 35064 | Columns: 18

--- Missing values check ---
Contains missing values? TRUE
Total missing values in dataset: 7271
```

Task 2 : Understand NA, NULL, and NaN

```
===== TASK 2: NA vs NULL vs NaN =====
```

```
NA example -> temperature: 28 30 NA 32  
is.na(temperature): FALSE FALSE TRUE FALSE
```

```
NULL example -> missing_object is NULL  
is.null(missing_object): TRUE
```

```
NaN example -> undefined_value (0/0): NaN  
is.nan(undefined_value): TRUE
```

Task 3 : Create a Missing-Value Summary Function

```
===== TASK 3: Missing Value Summary =====
```

	Variable	Total_Records	Missing_Values	Missing_Percentage
1	PM2.5	35064	925	2.64
2	PM10	35064	718	2.05
3	SO2	35064	935	2.67
4	NO2	35064	1023	2.92
5	TEMP	35064	20	0.06
6	WSPM	35064	14	0.04
7	wd	35064	81	0.23

Task 4 : Identify Invalid Numerical Results

```
===== TASK 4: pollution_ratio =====
```

```
NA count:      934
```

```
NaN count:      0
```

```
Infinite count: 0
```

```
After replacing NaN/Inf with NA -> NA count: 934
```

Task 5 : Handle Missing Numerical Values Using a Loop

===== TASK 5: Loop-based Numerical Cleaning =====

```
Variable: PM2.5
  Missing before: 925
  Median used:    58
  Missing after:  0

Variable: PM10
  Missing before: 718
  Median used:    87
  Missing after:  0

Variable: SO2
  Missing before: 935
  Median used:    9
  Missing after:  0

Variable: NO2
  Missing before: 1023
  Median used:    53
  Missing after:  0

Variable: TEMP
  Missing before: 20
  Median used:    14.5
  Missing after:  0

Variable: WSPM
  Missing before: 14
  Median used:    1.4
  Missing after:  0
```

Task 6: Categorical Cleaning (wd)

Task 7 : `clean_variable()` with `tryCatch`

===== TASK 6: Categorical Cleaning (wd) =====

Mode of wd: NE

Missing before: 81 | Missing after: 0

===== TASK 7: clean_variable() with tryCatch =====

-- Valid numeric variable --

Returned vector length: 35064

-- Non-existent variable --

Could not clean 'PM100': Variable 'PM100' does not exist in the dataset.

-- Categorical variable passed by mistake --

Could not clean 'wd': 'wd' is not a numerical variable.

Task 8 : Comparison Table

===== TASK 8: Comparison Table =====

	Variable	Missing_Before	Missing_After	Values_Replaced
1	PM2.5	925	0	925
2	PM10	718	0	718
3	SO2	935	0	935
4	NO2	1023	0	1023
5	TEMP	20	0	20
6	WSPM	14	0	14
7	wd	81	0	81

Interpretation: All selected variables show 0 missing values after cleaning, confirming median imputation (numeric columns) and mode imputation (categorical column 'wd') successfully handled the missing data.

Task 9 : Visualization

